STUDENT PRACTICE QUESTION BANK

Covers All Lecture Materials: Introduction, Sensors I, II & III |

PART I: MULTIPLE CHOICE QUESTIONS (80 Questions)

Instructions: Choose the single best answer for each question. Write your answers in the answer table provided before each set of 10 questions.

Set A — Sensor Fundamentals & Performance Terminology

Question	1	2	3	4	5	6	7	8	9	10
Answer										

Q1. A sensor is defined as an element that:
A) Converts only mechanical energy to electrical energy
B) Produces a signal relating to the quantity being measured
C) Amplifies and filters a transducer output signal
D) Stores measurement data for post-processing

Q2. Which of the following best describes a digital sensor?
A) One that provides a continuously varying analogue output proportional to the measurand
B) One that requires a DC power supply only
C) One that gives a sequence of on/off signals whose value represents the measurand
D) One that converts light intensity to a voltage

Q3. A smart sensor differs from a conventional sensor because it:
A) Uses only mechanical components for measurement
B) Combines the sensor, signal conditioning, and a microprocessor in a single package
C) Requires no power supply for operation
D) Can only measure a single physical quantity

Q4. Hysteresis error is defined as:
A) The time delay between input change and output settling
B) The maximum difference in output for the same input reached by increasing versus decreasing values
C) The deviation of the actual response from the assumed straight-line response
D) The variation in output caused by ambient temperature fluctuations

Q5. A transducer has a full-range output of 40 mV and a non-linearity of ±0.5%. What is the maximum non-linearity error in the output?
A) ±0.02 mV
B) ±0.2 mV
C) ±2.0 mV
D) ±5.0 mV

Q6. The time constant of a sensor is defined as the time required for the output to reach:
A) 50% of the steady-state value

B) 63.2% of the steady-state value
C) 90% of the steady-state value
D) 95% of the steady-state value

Q7. Rise time in the dynamic characteristics of a sensor is typically defined as the time for the output to rise from:
A) 0% to 100% of the steady-state value
B) 5% to 95% of the steady-state value
C) 10% to 90% (or 95%) of the steady-state value
D) 50% to 100% of the steady-state value

Q8. Settling time is defined as the time for the output to settle to within:
A) 50% of the steady-state value
B) 95% of the steady-state value
C) A specified small percentage (e.g. 2%) of the steady-state value
D) 63.2% of the steady-state value

Q9. Static characteristics of a transducer describe:
A) The transient response between input change and output settling
B) The behaviour of the sensor under ramp input conditions only
C) The values given when steady-state conditions occur
D) The frequency response to sinusoidal inputs

Q10. For a strain gauge pressure transducer (range 70–1000 kPa, full-range output 40 mV), a pressure of 750 kPa gives an output of approximately:
A) 20 mV
B) 30 mV
C) 28.7 mV
D) 35 mV

Set B — Displacement, Proximity & Position Sensors

Question	11	12	13	14	15	16	17	18	19	20
Answer										

Q11. Proximity sensors are distinguished from other displacement sensors because they:
A) Provide high-resolution continuous displacement measurements
B) Give on/off outputs when an object enters a defined critical distance
C) Use laser interferometry for sub-micron accuracy
D) Measure the velocity of a moving object directly

Q12. A potentiometer used as a linear displacement sensor converts:
A) Temperature into a resistance change
B) Force into a proportional current
C) Mechanical position or displacement into a proportional voltage output
D) Pressure into an angular rotation

Q13. A reed switch is classified as a:
A) Contact displacement sensor requiring physical contact
B) Non-contact proximity switch that closes when a magnet approaches
C) Capacitive sensor for fluid level detection
D) Piezoelectric pressure sensor

Q14. In washing machine water-level detection using a reed switch, the operating principle is:

A) A pressure transducer at the base of the drum senses hydrostatic pressure
B) A float carrying a magnet rises with the water level and triggers the reed switch at the set level
C) An ultrasonic sensor inside the drum measures the distance to the water surface
D) A resistive probe submerged in the water detects conductivity changes

Q15. An inductive proximity sensor can detect:
A) Both metallic and non-metallic objects equally
B) Metallic objects only, by the eddy-current loading effect on the coil oscillation
C) Only ferromagnetic materials above 100 °C
D) Clear transparent liquids at any distance

Q16. The resolution of an incremental optical encoder with 120 slots per revolution is:
A) 1.5°
B) 3°
C) 6°
D) 12°

Q17. An absolute encoder differs from an incremental encoder in that:
A) It counts pulses from a datum and resets on power-off
B) It gives the actual angular position at any instant without needing a datum reference
C) It uses a single-slot disc for position feedback
D) It requires an external counter to determine position

Q18. A transmissive (through-beam) photosensitive sensor detects an object by:
A) Reflecting light off the object surface back to the detector
B) The object physically interrupting a beam of light (or IR) between a separate transmitter and receiver
C) Measuring the heat radiation emitted by the target
D) Detecting the capacitance change near the object

Q19. The main advantage of ultrasonic proximity sensors over photoelectric sensors is:
A) They are completely immune to all electromagnetic interference
B) They can detect transparent or clear objects that may not reflect optical beams effectively
C) They operate only at very short ranges below 5 mm
D) They consume no electrical power in standby mode

Q20. When selecting a proximity sensor for a target distance greater than 6 m, the recommended sensor type is:
A) Inductive proximity sensor
B) Reed switch
C) Capacitive proximity sensor
D) Photoelectric sensor

Set C — LVDT, Capacitive Sensors & Strain Gauge Displacement

Question	21	22	23	24	25	26	27	28	29	30
Answer										

Q21. In an LVDT, the net output voltage is zero when the magnetic core is at the central position because:
A) The primary coil is disconnected at that position
B) Equal e.m.f.s are induced in both secondary coils, and since they are connected in opposition the net output is zero

C) The magnetic core blocks all flux at the centre
D) The two secondary coils are short-circuited at the centre

Q22. A phase-sensitive demodulator is used with the LVDT output because:
A) The raw primary coil signal is too large for direct measurement
B) The same output amplitude is produced for symmetrically opposite displacements, and the phase difference (180°) must be used to distinguish direction
C) The secondary coils generate large noise that must be filtered
D) The primary AC supply must be converted to DC first

Q23. A strain-gauge displacement sensor using cantilever/ring flexible elements typically covers linear displacements of:
A) 0.001 to 0.1 mm
B) 1 to 30 mm
C) 100 to 500 mm
D) Over 1 metre

Q24. A capacitive accelerometer in an automobile airbag system detects a crash by:
A) Measuring the temperature spike during impact
B) Detecting the differential capacitance change between fixed substrate plates and moving frame plates during deceleration
C) Using a piezoelectric crystal voltage generated by the impact force
D) Monitoring the inductance change of a coil mounted on the bumper

Q25. In the push-pull displacement sensor, using two potentiometers in opposition gives:
A) Double sensitivity because both outputs add together
B) A differential output where one resistance increases as the other decreases, improving linearity and doubling sensitivity
C) A digital output pulse train proportional to displacement
D) Temperature compensation by cancelling thermal drift

Q26. A capacitive sensor used as a humidity sensor detects moisture because:
A) Water droplets physically bridge the two capacitor plates
B) One permeable capacitor plate absorbs water, changing the dielectric constant and thus the capacitance
C) Water increases the plate area of the capacitor
D) Humidity changes the supply voltage to the sensor

Q27. An RVDT (Rotary Variable Differential Transformer) is used to measure:
A) Linear displacement up to 500 mm
B) Angular rotation (rotational displacement)
C) Fluid flow velocity
D) Thermal expansion of metals

Q28. The inductive proximity sensor detection circuit triggers the output when:
A) The oscillating magnetic field increases to a maximum
B) The eddy-current loading of a nearby metallic object reduces the oscillating magnetic field current to a sufficient level
C) The target comes within 1 mm of the sensor face only
D) A non-metallic object approaches the coil

Q29. The inner track of an incremental optical encoder disc with one hole is used to:
A) Determine the direction of rotation
B) Set the resolution of the encoder
C) Provide a home/datum (index) position reference
D) Count the total number of revolutions

Q30. The LVDT is described as an ideal displacement sensor for many applications because:

A) It produces very high output voltages requiring no amplification
B) There is no mechanical contact between the moving core and the coil assembly, giving virtually infinite mechanical life and no friction
C) Its output is always a digital signal
D) It can only measure angular displacement

Set D — Magnetic Sensors & Velocity Measurement

Question	31	32	33	34	35	36	37	38	39	40
Answer										

Q31. In the Hall effect formula V = K_H × (BI/t), the variable "t" represents:
A) The time duration of measurement
B) The operating temperature
C) The thickness of the conductive plate
D) The transmittance of the sensor material

Q32. When a constant current source is used with a Hall effect sensor, the Hall voltage becomes a direct measure of:
A) The electric current magnitude
B) The magnetic flux density (B)
C) The electric field intensity
D) The sensor temperature

Q33. The Schmitt trigger in a digital-output Hall effect sensor serves to:
A) Convert the Hall voltage from analogue to digital
B) Provide hysteresis to prevent false triggering caused by minor fluctuations in the input magnetic field
C) Regulate the supply voltage to the Hall element
D) Amplify the output to drive high-power loads

Q34. A magneto-resistive sensor differs from a Hall effect sensor in that it:
A) Produces a voltage change proportional to field strength
B) Gives a resistance change with the field applied perpendicular to the sensor plane, and its output is independent of magnet polarity
C) Can measure both field strength and polarity simultaneously
D) Uses a Wheatstone bridge for force measurement

Q35. Magneto-resistive sensors are typically used for:
A) Precise measurement of magnetic field strength
B) Position detection – indicating whether an element (e.g. gear tooth, door) is present or absent
C) Measuring angular velocity with high accuracy
D) Controlling motor drive current

Q36. The variable reluctance tachogenerator produces an output e.m.f. by:
A) A Hall effect chip mounted on the rotating shaft
B) A toothed ferromagnetic wheel rotating past a pick-up coil wound on a permanent magnet, periodically changing the air gap and hence the flux linkage
C) A capacitive sensor measuring shaft rotation
D) A reed switch activated by teeth on the wheel

Q37. For a tachogenerator with a wheel of n teeth rotating at angular velocity ω, the peak (maximum) e.m.f. is proportional to:
A) n alone

B) ω alone
C) n × ω (and the coil turns N and flux amplitude Φ_a)
D) 1 / (n × ω)

Q38. An alternative to using the peak e.m.f. of a tachogenerator to measure angular velocity is to:
A) Measure the DC component of the output
B) Use a pulse-shaping signal conditioner to convert the output into a pulse train and count pulses over a fixed time interval
C) Integrate the e.m.f. over one revolution
D) Measure the phase angle of the AC output

Q39. A rotary magnetic encoder (Hall effect type) achieves position detection by using:
A) A transparent disc with optical slots and photodetectors
B) A magnetic disc with alternating North and South poles that trigger a Hall effect sensor as the disc rotates
C) A reed switch activated by a single magnet on the shaft
D) An LVDT core attached to the rotating shaft

Q40. An AC tachogenerator (rotor-type) measures angular velocity using:
A) The frequency or amplitude of the AC e.m.f. induced in a rotor coil rotating in a stationary magnetic field
B) A strain gauge attached to the rotor shaft
C) The capacitance change between rotor and stator plates
D) The Hall voltage produced in the rotating conductor

Set E — Force, Pressure & Piezoelectric Sensors

Question	41	42	43	44	45	46	47	48	49	50
Answer										

Q41. A strain gauge load cell measures force by:
A) Measuring the thermal expansion of a metal column under load
B) Monitoring the electrical resistance change of strain gauges bonded to an elastic element deformed by the applied force
C) Using a capacitive plate to detect bending of a beam
D) Measuring the magnetic flux change in a ferromagnetic member

Q42. The signal conditioning circuit of a strain gauge load cell must compensate for temperature because:
A) Strain gauges only work above 20 °C
B) Temperature also causes resistance changes in the strain gauges, which would be misinterpreted as force-induced strain
C) The elastic element loses stiffness rapidly with temperature
D) The supply voltage changes with temperature

Q43. Absolute pressure is defined as:
A) Pressure measured relative to atmospheric (barometric) pressure
B) The difference in pressure between two points in a system
C) Pressure measured relative to zero pressure (perfect vacuum)
D) The maximum safe pressure rating of a vessel

Q44. Corrugations added to a diaphragm pressure sensor result in:
A) Reduced mechanical sensitivity
B) Greater deflection sensitivity for a given pressure difference, due to increased flexibility

C) A higher minimum detectable pressure
D) Digital output from the diaphragm

Q45. Capsules in pressure sensing offer greater sensitivity than a single diaphragm because:
A) They use a thicker material to withstand higher pressure
B) A capsule is essentially two corrugated diaphragms combined, giving cumulative deflection sensitivity
C) They incorporate electronic amplifiers internally
D) They remove the need for strain gauges

Q46. A piezoelectric accelerometer measures vibration acceleration because:
A) A capacitive plate detects the movement of a seismic mass
B) The vibration transmits acceleration to a seismic mass which applies a proportional force to a piezoelectric crystal, generating a charge/voltage proportional to acceleration
C) A bimetallic strip bends proportionally to vibration amplitude
D) A strain gauge on a cantilever detects the seismic mass deflection

Q47. The reverse piezoelectric effect is used to:
A) Convert mechanical vibration into an electrical voltage
B) Generate ultrasonic waves when a high-frequency alternating voltage is applied to the piezoelectric crystal
C) Measure static forces using a crystal
D) Produce a Hall voltage in a magnetic field

Q48. A Force-Sensitive Resistor (FSR) operates on the principle that:
A) Its resistance increases when more force is applied to its surface
B) The applied force presses the conductive film against the resistive layer; greater contact area means lower sensor resistance
C) The piezoelectric material generates a charge proportional to force
D) A strain gauge bonded to a flexible film changes resistance under force

Q49. For simple force-to-voltage conversion, the FSR is connected:
A) In series with a current source to give a proportional current output
B) In a voltage divider with a fixed resistor of similar resistance; the output is taken across the fixed resistor
C) Directly to a microcontroller digital pin without any resistor
D) In parallel with a capacitor to integrate the force signal

Q50. The main drawback of the FSR sensor is:
A) Its inability to work below 10 N applied force
B) The plastic spacer's slow re-inflation after compression, causing a slow drift of approximately 5% back to the initial resistance value over several seconds
C) Its resistance being too high to measure with standard circuits
D) Its incompatibility with microcontroller analogue inputs

Set F — Liquid Level & Temperature Sensors

Question	51	52	53	54	55	56	57	58	59	60
Answer										

Q51. An indirect method of liquid level measurement using a load cell works by:
A) Measuring the pressure at the base of the vessel
B) Monitoring the total weight of the vessel plus liquid; changes in liquid height produce proportional weight changes ($W = Ah\rho g$)

C) Detecting the capacitance change between two vertical plates in the vessel
D) Using a float to move an LVDT core

Q52. In a float-based liquid level sensor, the float displacement is converted to a voltage output by:
A) A strain gauge attached directly to the float
B) A lever arm that rotates a potentiometer wiper or displaces an LVDT core proportionally to the liquid height
C) A differential pressure cell beneath the float
D) A thermistor mounted on the float

Q53. For a closed vessel, liquid level measurement using a differential pressure cell monitors:
A) The pressure at the top of the liquid only
B) The difference in pressure between the base of the vessel and the gas/air above the liquid surface
C) The absolute pressure at the base relative to perfect vacuum
D) The weight of the vessel using load cells on the base

Q54. A bimetallic strip thermostat operates because:
A) Both metal strips have identical thermal expansion coefficients
B) The two bonded metals have different thermal expansion coefficients, so the composite strip bends on temperature change, making or breaking a contact
C) One metal becomes ferromagnetic only at elevated temperatures
D) The strip changes electrical resistance proportionally to temperature

Q55. The RTD (Resistance Temperature Detector) formula $R_t = R_0(1 + at)$ indicates that over a limited range:
A) Resistance decreases non-linearly with increasing temperature
B) Resistance increases approximately linearly with temperature, where a is the temperature coefficient
C) Resistance is inversely proportional to temperature
D) Resistance is independent of the material used

Q56. Which metal among Nickel, Copper, and Platinum gives the highest R_t/R_0 ratio at elevated temperatures (according to RTD characteristic curves)?
A) Platinum
B) Copper
C) Nickel
D) All three are identical above 400 °C

Q57. A key disadvantage of NTC thermistors compared to RTDs is:
A) They are larger and less robust than RTDs
B) Their resistance-temperature relationship is highly non-linear, requiring linearisation
C) They respond more slowly to temperature changes
D) They are more expensive than platinum RTDs

Q58. PTC thermistors are used as motor overload protection because:
A) Their resistance decreases sharply when the motor overheats, cutting the current
B) Their resistance increases dramatically over a small temperature range, acting as an effective thermal cutoff switch
C) They generate an e.m.f. proportional to motor current
D) They act as varistors, clamping the voltage during overloads

Q59. A thermocouple generates an e.m.f. because:
A) One metal expands more thermally than the other
B) Two dissimilar metals joined at a junction develop a potential difference that depends on the junction temperature
C) One metal becomes magnetic and the other does not

D) Electrons are blocked at the junction, creating a charge difference

Q60. The thermocouple output e.m.f. equation $E = at + bt^2$ implies:
A) The output is perfectly linear with temperature t
B) There is a slight non-linearity due to the bt^2 quadratic term
C) The output decreases with increasing temperature
D) The output is independent of the metal types used

Set G — Light Sensors, Semiconductor Temp & Switches

Question	71	72	73	74	75	76	77	78	79	80
Answer										

Q71. A photodiode used as a light sensor is connected in:
A) Forward bias, where forward current increases with illumination
B) Reverse bias, where the normally negligible reverse (dark) current increases proportionally with light intensity
C) Alternating bias to allow both positive and negative light signals
D) Zero bias with a pull-up resistor only

Q72. The dark current in a photodiode refers to:
A) The photocurrent generated under maximum illumination
B) The very small reverse current flowing when no light is incident on the junction
C) The current generated by thermal noise at elevated temperatures
D) The current that flows under forward-bias conditions

Q73. A phototransistor connected in Darlington configuration provides:
A) Lower sensitivity with reduced noise
B) A much higher current gain, resulting in significantly greater collector current for a given light intensity
C) A purely digital (on/off) output only
D) Temperature compensation for the base-emitter junction

Q74. In a thermodiode (e.g. LM35), when constant current flows through the p-n junction, the measured electrical quantity that varies with temperature is:
A) The forward current
B) The reverse breakdown voltage
C) The voltage across the junction, which has a linear dependence on temperature
D) The capacitance of the depletion layer

Q75. A thermodiode temperature sensor has an advantage over a thermistor because:
A) It has a lower cost of manufacture
B) It is available in larger physical sizes
C) It has a linear response with temperature, unlike the non-linear thermistor
D) It can operate at higher temperatures than any thermistor

Q76. In a thermotransistor (e.g. for temperature sensing), the temperature-sensitive quantity that is measured is:
A) The collector supply voltage
B) The emitter current under constant base voltage
C) The base-emitter junction voltage, which is temperature-dependent
D) The frequency of oscillation of the transistor circuit

Q77. Mechanical switches are specified by "poles" and "throws." Poles are defined as:

A) The number of separate circuits that can be controlled by one switching action
B) The number of independent throw positions available to each contact
C) The maximum current rating of the switch contacts
D) The physical number of terminals on the switch body

Q78. In the standard switch interface circuit (pull-up resistor circuit), when the switch is CLOSED, the output voltage is:
A) Equal to the supply voltage
B) Approximately zero (since Rs ≈ 0 when closed, and output is across Rs)
C) Half the supply voltage
D) Negative

Q79. A limit switch is best described as:
A) An electronic sensor measuring precise linear displacement continuously
B) A mechanical switch actuated by the physical contact of a moving object to indicate when it has reached its displacement limit
C) A capacitive sensor that detects the proximity of non-metallic objects
D) A current-controlled switch for motor speed regulation

Q80. The silicon diaphragm pressure sensor is used in automobile electronic systems to monitor:
A) Tyre pressure from the outside
B) Engine coolant temperature
C) Inlet manifold air pressure (MAP sensor)
D) Fuel injection timing

Set H — Introduction to Mechatronics: Systems, Actuators & Microcontrollers

Question	61	62	63	64	65	66	67	68	69	70
Answer										

Q61. The term "mechatronics" was coined by which company in the 1970s?
A) Toyota Motor Corporation
B) Yasakawa Electric Company
C) Mitsubishi Heavy Industries
D) Sony Corporation

Q62. Mechatronics is best defined as:
A) The study of mechanical design principles for robots
B) The application of electronics and computer technology to control the motions of mechanical systems
C) A branch of electronics focused on circuit design only
D) The use of hydraulic systems in manufacturing

Q63. Which of the following lists correctly identifies the four key elements of a mechatronics system?
A) Sensors, resistors, capacitors, and inductors
B) Sensors, actuators, controllers, and mechanical/electrical components
C) Microprocessors, displays, motors, and gears
D) Software, hardware, firmware, and databases

Q64. Control is defined as the process of:
A) Designing mechanical components for a system
B) Measuring physical quantities with high accuracy
C) Altering, manually or automatically, the performance of a system to a desired outcome

D) Converting one form of energy into another

Q65. A transducer is defined as a device that:
A) Measures a physical quantity and stores the data
B) Converts one form of energy into another form of energy
C) Amplifies electrical signals for further processing
D) Controls the speed of a motor using feedback

Q66. An actuator is a component responsible for:
A) Measuring physical variables and converting them to electrical signals
B) Moving or controlling a mechanism by converting electrical/hydraulic/pneumatic power into physical motion
C) Processing sensor data inside a microcontroller
D) Storing program instructions for a controller

Q67. Which of the following is NOT one of the four main types of actuators?
A) Hydraulic
B) Pneumatic
C) Electric
D) Optical

Q68. A microcontroller integrates which three major components on a single chip?
A) Sensor, actuator, and display
B) Microprocessor (MPU), memory, and I/O ports
C) ADC, DAC, and power supply
D) Clock, bus, and interrupt controller only

Q69. Arduino is described as an open-source electronics platform because it:
A) Is manufactured without patents and sold freely
B) Integrates hardware and software to allow reading inputs and producing outputs, with boards that can be programmed with open instructions
C) Uses only open-source operating systems
D) Is powered exclusively by solar energy

Q70. According to the key elements of mechatronics study, which of the following is NOT one of the five specialty areas?
A) Physical Systems Modeling
B) Sensors and Actuators
C) Fluid Mechanics and Thermodynamics
D) Software and Data Acquisition

PART II: FILL-IN-THE-BLANK QUESTIONS (30 Questions)

Instructions: Complete each sentence by filling in the blanks. Write the answers on the lines provided.

Q1. The time constant of a sensor is the time for its output to reach ________ % of the final steady-state value after a step input is applied.
Answer: ____

Q2. The response time of a sensor is defined as the time elapsed after a step input is applied until the output reaches ________ % of the input value.
Answer: ____

Q3. The RTD resistance formula is R_t = R_0(1 + ________), where the symbol 'a' represents the ________ coefficient of the metal.
Answer: ____

Q4. The thermocouple e.m.f. equation is E = _______, showing that the output has a slight _______ component.

Answer: ____

Q5. A magneto-resistive sensor gives a _______ change as its output, whereas a Hall effect sensor gives a _______ change.

Answer: ____

Q6. In the LVDT, the central coil is the _______ coil, and the two identical outer coils are connected so that their outputs _______ each other.

Answer: ____

Q7. A reed switch consists of two magnetic contacts sealed in a _______ tube. It closes when a _______ is brought close to it.

Answer: ____

Q8. An incremental optical encoder with N slots per revolution has an angular resolution of _______ degrees.

Answer: ____

Q9. For the NTC thermistor, resistance _______ as temperature increases, and this change is highly _______.

Answer: ____

Q10. The HC-SR04 ultrasonic sensor has _______ pins. The formula to calculate distance is: Distance = (duration / _______) × _______.

Answer: ____

Q11. In the Hall effect formula V = K_H × (BI/t), B is the _______, I is the _______, and t is the _______ of the conductive plate.

Answer: ____

Q12. The push-pull displacement sensor uses two potentiometers in _______ so that as one resistance increases the other _______, improving sensitivity.

Answer: ____

Q13. A piezoelectric accelerometer uses a _______ mass to transmit acceleration forces to a piezoelectric crystal, which generates a proportional _______.

Answer: ____

Q14. The FSR (Force-Sensitive Resistor) resistance _______ as applied force increases. For force-to-voltage conversion it is placed in a _______ circuit.

Answer: ____

Q15. In mechanical switch specification, "poles" are the number of separate _______ that one switch action can control, and "throws" are the number of _______ per pole.

Answer: ____

Q16. Gauge pressure is measured relative to _______ pressure, whereas absolute pressure is measured relative to _______.

Answer: ____

Q17. A bellows pressure element can be combined with an _______ to produce a pressure sensor with an electrical output.

Answer: ____

Q18. The variable reluctance tachogenerator produces an AC e.m.f. whose _______ is proportional to the angular velocity of the toothed wheel.

Answer: ____

Q19. In a bimetallic strip, the metal with the _______ thermal expansion coefficient ends up on the outside of the curve when heated.

Answer: ____

Q20. A photodiode is operated in _______ bias. The current that flows in the absence of light is called the _______ current.

Answer: ____

Q21. Smart sensors have the ability to self-_______, compensate for non-_______, and perform self-_______ of faults.

Answer: ____

Q22. An inductive proximity sensor detects _______ objects only, by the effect of _______ currents reducing the coil's oscillating magnetic field.

Answer: ____

Q23. The settling time is the time for the output to settle within a specified _______ (e.g. _______ %) of the steady-state value.

Answer: ____

Q24. The non-linearity error is generally quoted as a percentage of the _______ range _______, which represents the maximum output of the sensor.

Answer: ____

Q25. In the digital Hall effect sensor, the _______ prevents false triggering, while the _______ stabilises the supply voltage to the Hall element.

Answer: ____

Q26. The term "mechatronics" was coined by _______ Electric Company in the _______ to refer to the use of electronics in mechanical control.

Answer: ____

Q27. The four key elements of a mechatronics system are: _______, _______, _______, and mechanical/electrical components.

Answer: ____

Q28. An actuator converts an electrical signal into a _______ quantity. The four main types of actuators are: _______, _______, Electric, and _______.

Answer: ____

Q29. A microcontroller integrates three major components on a single chip: _______ (MPU), _______, and _______ ports.

Answer: ____

Q30. The five specialty areas in mechatronics study are: Physical Systems _______, Sensors and Actuators, Signals and _______, Computers and Logic _______, and Software and Data _______.

Answer: ____

PART III: DRAWING & APPLICATION QUESTIONS (23 Questions)

Instructions: For each question, draw a clear, labelled diagram or schematic in the space provided. Label all key components.

Q1. Draw the construction of a linear potentiometer used as a displacement sensor, showing the resistive track, wiper, and the output voltage circuit. Label all components.

Q2. Draw the internal construction of an LVDT (Linear Variable Differential Transformer), clearly labelling the primary coil, two secondary coils, magnetic core, and insulated tube. Add the voltage–displacement output curve.

Q3. Draw the circuit diagram of a push-pull displacement sensor using two potentiometers and show how the differential voltage output is obtained.

Q4. Draw a schematic diagram showing how a reed switch and float system are used to detect the maximum water level in a washing machine drum. Label the magnet, float, switch, and circuit connections.

Q5. Draw the block diagram of a smart sensor showing all internal components: sensing element, signal conditioning, microprocessor/ADC, and output interface.

Q6. Draw the construction of a variable reluctance tachogenerator (tacho-generator), showing the toothed ferromagnetic wheel, permanent magnet, pick-up coil, and the shape of the output waveform versus rotation.

Q7. Draw the Hall effect sensor (analogue output type) block diagram, showing: Hall element, voltage regulator, amplifier, VCC, GND, and output terminals. Sketch the output voltage vs. input magnetic field characteristic.

Q8. Draw the digital output Hall effect sensor block diagram including: Hall element, differential amplifier, Schmitt trigger, voltage regulator, and output. Sketch the ON/OFF output vs. magnetic field curve showing hysteresis.

Q9. Draw the cross-section of a diaphragm-based fluid pressure sensor (corrugated type) and show where the strain gauge is placed. Label: diaphragm, corrugations, strain gauge, pressure inlet, and fixed housing.

Q10. Draw the construction of a piezoelectric accelerometer. Label: seismic mass, piezoelectric crystal, preload spring, damper, conductive coating, accelerometer housing, and vibrating base.

Q11. Draw the force-sensitive resistor (FSR) voltage divider circuit used for force-to-voltage conversion with a 100 kΩ fixed resistor and a +5 V supply. Label the output node and indicate what happens to Vout when force increases.

Q12. Draw a schematic of the liquid level measurement system for a closed vessel using a differential pressure cell (DP), showing both connection points and labelling the DP cell, vessel, liquid surface, and gas space.

Q13. Draw the NTC thermistor voltage divider circuit connected to the analogue input of an Arduino UNO. Label all components, supply voltage, and output pin.

Q14. Draw the LM35 thermodiode temperature sensor circuit connected to an Arduino UNO analogue input. Label VCC (+5 V), GND, and output connections.

Q15. Draw the MAX6675 thermocouple module connected to an Arduino UNO, clearly showing the three SPI signal connections: SCK (Pin 6), CS (Pin 5), and SO (Pin 4), as well as power connections.

Q16. Draw the construction of an incremental optical encoder disc, showing: the inner single-hole index track, two outer offset tracks, LED source, and photodetector. Indicate how direction of rotation is determined.

Q17. Draw a strain gauge load cell (50 kg capacity) as a secondary sensor for measuring the level of liquid in a cylindrical vessel. Show how the load cell supports the vessel and how its output relates to liquid level.

Q18. Draw the block diagram of the LVDT signal conditioning circuit that converts the raw AC differential output to a DC voltage unique to each displacement. Include: LVDT coils, phase-sensitive demodulator, and low-pass filter.

Q19. Draw the circuit diagram for connecting a Hall effect sensor module (A3144 type with AO, DO, GND, VCC pins) to an Arduino UNO, showing both the digital and analogue output connections.

Q20. Draw a capacitive liquid level sensor arrangement showing two vertical parallel plate electrodes inside a vessel. Explain how rising liquid level between the plates changes the capacitance.

Q21. Draw the block diagram of a complete closed-loop mechatronics control system for automobile active suspension. Label the four key elements: sensor (road condition sensor), controller (ECU), actuator (hydraulic/pneumatic strut), and mechanical system (suspension). Show the feedback path.

Q22. Draw the internal architecture block diagram of a microcontroller, showing its three main integrated components: Microprocessor (MPU), Memory (RAM/ROM), and I/O Ports. Add arrows to show data flow between components and label the external connections to sensors and actuators.

Q23. Draw a labelled diagram of the Arduino UNO board identifying: digital I/O pins, analogue input pins (A0–A5), power pins (5V, 3.3V, GND), USB connector, reset button, and the ATmega328P microcontroller chip.

ANSWER KEY (Multiple Choice — Part I)

Note: This section is for instructor reference only.

Q1: B	Q2: C	Q3: B	Q4: B	Q5: B	Q6: B	Q7: C	Q8: C	Q9: C	Q10: C
Q11: B	Q12: C	Q13: B	Q14: B	Q15: B	Q16: B	Q17: B	Q18: B	Q19: B	Q20: D
Q21: B	Q22: B	Q23: B	Q24: B	Q25: B	Q26: B	Q27: B	Q28: B	Q29: C	Q30: B
Q31: C	Q32: B	Q33: B	Q34: B	Q35: B	Q36: B	Q37: C	Q38: B	Q39: B	Q40: A
Q41: B	Q42: B	Q43: C	Q44: B	Q45: B	Q46: B	Q47: B	Q48: B	Q49: B	Q50: B
Q51: B	Q52: B	Q53: B	Q54: B	Q55: B	Q56: C	Q57: B	Q58: B	Q59: B	Q60: B
Q61: B	Q62: B	Q63: B	Q64: C	Q65: B	Q66: B	Q67: D	Q68: B	Q69: B	Q70: C
Q71: B	Q72: B	Q73: B	Q74: C	Q75: C	Q76: C	Q77: A	Q78: B	Q79: B	Q80: C

Fill-in-the-Blank Answer Key (Part II)

Q1: 63.2

Q2: 95

Q3: at ; temperature

Q4: $at + bt^2$; quadratic (non-linear)

Q5: resistance ; voltage

Q6: primary ; oppose

Q7: glass ; magnet

Q8: 360 / N

Q9: decreases ; non-linear
Q10: 4 ; 2 ; speed of sound (≈ 0.034 cm/μs)
Q11: magnetic flux density ; current ; thickness
Q12: opposition (differential) ; decreases
Q13: seismic ; potential difference (voltage)
Q14: decreases ; voltage divider
Q15: circuits ; contacts (positions)
Q16: atmospheric (barometric) ; zero (perfect vacuum)
Q17: LVDT
Q18: amplitude (maximum value)
Q19: higher
Q20: reverse ; dark
Q21: calibrate ; linearities ; diagnosis
Q22: metallic ; eddy
Q23: percentage ; 2
Q24: full ; output
Q25: Schmitt trigger ; voltage regulator
Q26: Yasakawa ; 1970s
Q27: Sensors ; Actuators ; Controllers
Q28: physical ; Hydraulic ; Pneumatic ; Mechanical
Q29: Microprocessor ; Memory ; I/O
Q30: Modeling ; Systems ; Systems ; Acquisition